

Name: Kumaraswamy S
USN: 1RV24MC058

Program 3: Code a Dockerized Python Flask or Node.js Application

Project structure:

```
.
├── Dockerfile
├── index.js
├── package.json
├── node_modules
│   └── .....
└── package-lock.json
```

STEP 1: Create a Dockerfile and add the following content.

```
FROM node:latest
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
CMD ["npm", "start"]
```

STEP 1.2: Create a index.js and add the following content.

```
import express from "express";
const app = express();
const PORT = 3000;
app.use(express.json());
app.get("/", (req, res) => {
  res.send("docker is runnig");
});
app.listen(PORT, "0.0.0.0", () => console.log(`Server running on port ${PORT}`));
```

STEP 1.2: Create a node project

```
npm init -y
npm install express
```

STEP 2: Execute the Docker Build command on the Terminal to build the Docker image and then execute the Docker run command specifying the Port Number for the container.

➤ **sudo docker build -t node1 .**

```
> sudo docker build -t node1 .
[sudo] password for kumars:
Sorry, try again.
[sudo] password for kumars:
[+] Building 18.8s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 153B                                0.0s
=> [internal] load metadata for docker.io/library/node:latest     4.2s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:latest@sha256:a84dedcbbcf651db49178dd876bad1ea 13.3s
=> => resolve docker.io/library/node:latest@sha256:a84dedcbbcf651db49178dd876bad1ea6 0.0s
=> => sha256:876cf39ddf32fd345a43c42d3dc000bfe59ab966bbcb3f957a98b90 6.74kB / 6.74kB 0.0s
```

STEP 3: run the containers.

```
> sudo docker images
```

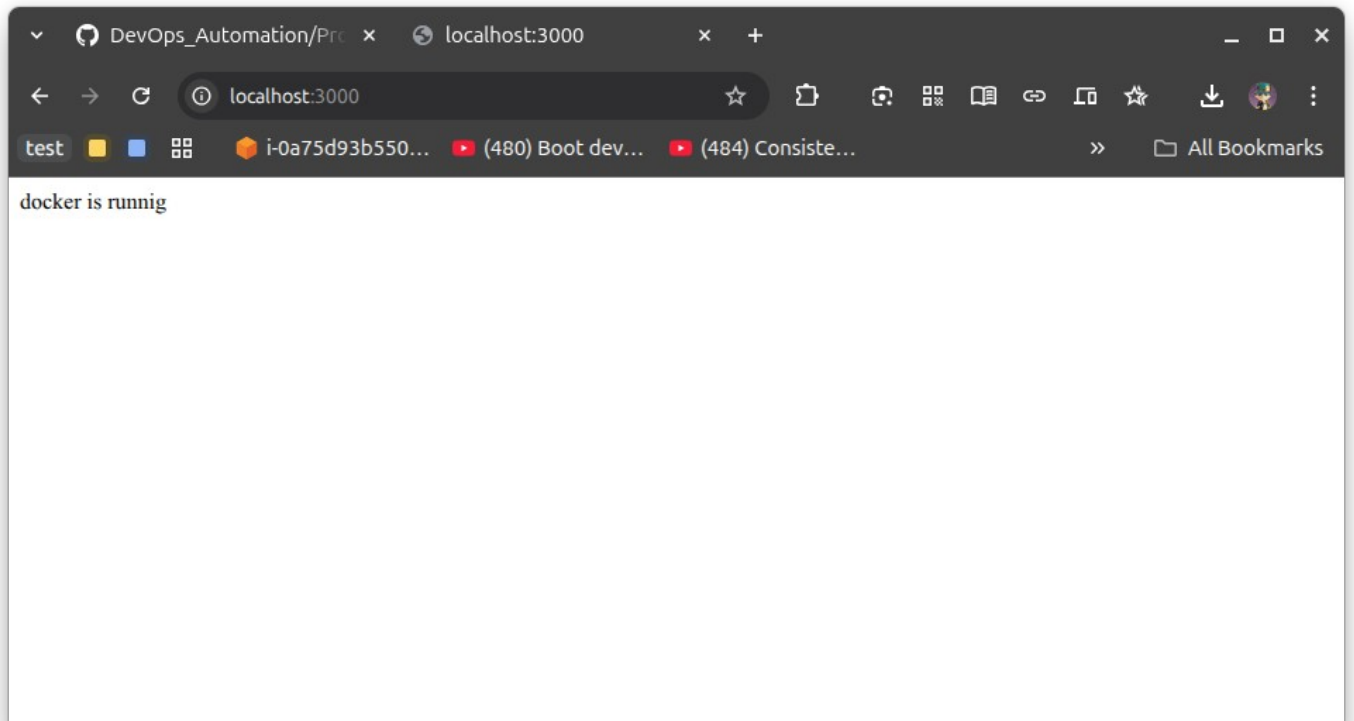
IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
node1:latest	9cf1a6b06e51	1.14GB	0B	

```
> sudo docker run -p 3000:3000 node1
```

```
> docker@1.0.0 start
```

```
> node index.js
```

Server running on port 3000



STEP 4: Stop and remove the container with the container number, remove the docker container and delete the Docker Image

› **sudo docker rmi node1**